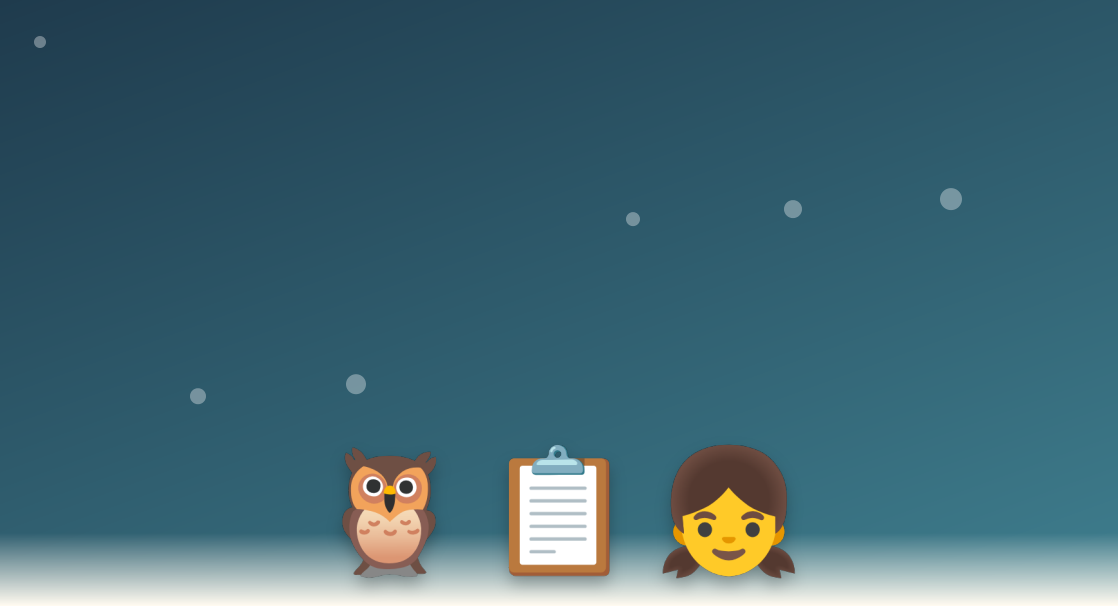


# The Great Kingdom Census

I PUC · Ch 2 · Biological Classification

Starring Bindu 👧 & Prof. BAIO 🦉

A BAIO storybook by Narendra Sir · Biology Lecturer



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# The Census Begins

Prof. BAIO pinned a giant chart on the lab wall. "Bindu, every living thing needs a home address. In 1969, **R.H. Whittaker** sorted all life into five kingdoms." Bindu read them aloud: **Monera**, **Protista**, **Fungi**, **Plantae** and **Animalia**. "Today," said the owl, "we visit them all!"

💡 REMEMBER

**Whittaker's five kingdoms are based on cell structure, body organisation, mode of nutrition, reproduction and phylogenetic relationships.**



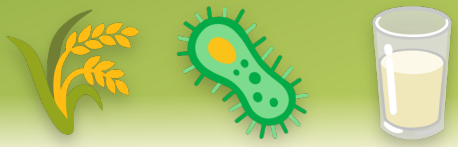
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# Kingdom Monera

The first stop was the kingdom of the tiniest. "Everyone here is a **prokaryote** — no nuclear membrane!" hooted Prof. BAIO. In a boiling hot spring lived the **thermoacidophiles**. In salty pans lived the **halophiles**. And inside a cow's stomach, the **methanogens** were busy making methane gas.

💡 REMEMBER

**Archaebacteria survive extreme places because of a special cell wall. Methanogens produce methane — biogas.**



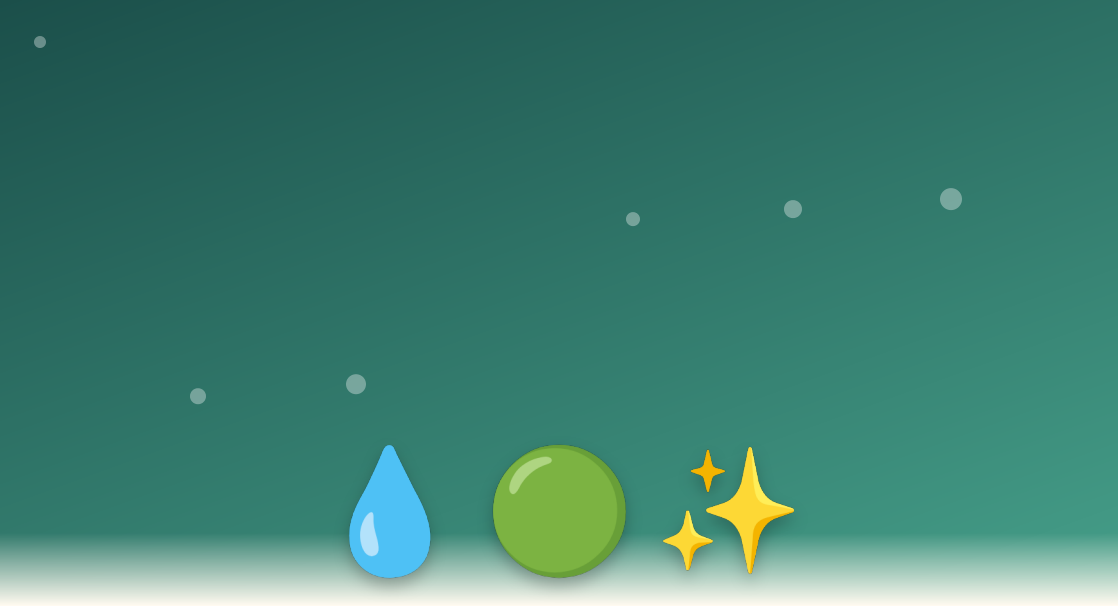


# The Useful Bacteria

Next came the **eubacteria**, the true bacteria. Green **cyanobacteria** like Nostoc and Anabaena waved from a paddy field — they fix nitrogen in special cells called **heterocysts**. Then the smallest of all, **Mycoplasma**, peeped out shyly. "I have no cell wall at all!"

💡 **REMEMBER**

**Mycoplasma has no cell wall and is the smallest living cell; it can survive without oxygen. Bacteria reproduce mainly by fission.**





# The Protist Pond

In a sparkling pond lived the single-celled eukaryotes of **Protista**. **Diatoms** glittered in glassy silica shells; their piled-up walls form **diatomaceous earth**. **Euglena** made food in sunlight but hunted in the dark. Far out at sea, **dinoflagellates** like Gonyaulax multiplied so fast that the water turned red — a red tide!

💡 **REMEMBER**

**Diatoms are the chief producers in the oceans. Euglena has a protein-rich pellicle instead of a cell wall.**





# The Protozoan Parade

Four kinds of **protozoans** marched past. Amoeba crawled on **pseudopodia**. Trypanosoma whipped its flagellum — it causes sleeping sickness. Paramecium rowed with tiny **cilia**. Last came **Plasmodium**, a sporozoan, who whispered, "I cause malaria."

💡 **REMEMBER**

**Protozoans: amoeboid (Amoeba), flagellated (Trypanosoma), ciliated (Paramecium) and sporozoans (Plasmodium).**





# The Mushroom Mansion

The Fungi kingdom smelled of bread and rain. Their walls were made of **chitin**, and their bodies were long threads called **hyphae** woven into a **mycelium**. Rhizopus grew on old bread, Puccinia gave wheat its rust, and Penicillium gave the world an antibiotic.

💡 **REMEMBER**

**Fungi are heterotrophs. Classes: Phycomycetes, Ascomycetes, Basidiomycetes and Deuteromycetes (the "imperfect fungi").**





# Not Quite Alive

In a dark corner hid three puzzles. A **virus** was just a nucleoprotein, lifeless outside a host — Ivanowsky discovered one in tobacco in 1892. A **viroid** was naked RNA with no protein coat. A **prion** was a wrongly folded protein that causes mad cow disease.

💡 **REMEMBER**

**Viruses are not placed in any of the five kingdoms. Viroids were discovered by T.O. Diener (1971) — potato spindle tuber disease.**





# The Best Friends

On a rock, an alga and a fungus waved together. "We are a **lichen**," said the alga. "I make the food — I am the **phycobiont**." "And I give shelter and water — I am the **mycobiont**," said the fungus. Prof. BAIO smiled: "Remember, lichens never grow where the air is polluted."

💡 **REMEMBER**

**Lichens are a mutual partnership of algae and fungi — they are good indicators of air pollution.**



KEY TERMS

# Words you met in this story

Tap a word to jump to its page. Can you explain each one?

R.H. Whittaker p.1 · Monera p.1 · Protista p.1 · Fungi p.1 · Plantae p.1 · Animalia p.1 · prokaryote p.2 · thermoacidophiles p.2 · halophiles p.2 · methanogens p.2 · eubacteria p.3 · cyanobacteria p.3 · heterocysts p.3 · Mycoplasma p.3 · Diatoms p.4 · diatomaceous earth p.4 · Euglena p.4 · dinoflagellates p.4 · protozoans p.5 · pseudopodia p.5 · cilia p.5 · Plasmodium p.5 · chitin p.6 · hyphae p.6 · mycelium p.6 · virus p.7 · viroid p.7 · prion p.7 · lichen p.8 · phycobiont p.8 · mycobiont p.8



QUICK QUIZ

# Do you remember?

**1. Who proposed the five kingdom classification?**

- Linnaeus
- R.H. Whittaker
- Darwin
- Haeckel

**2. Which organism has no cell wall?**

- Nostoc
- Mycoplasma
- Euglena
- Diatom

**3. Methanogens produce:**

- Oxygen
- Methane
- Nitrogen
- Ammonia

**4. The cell wall of fungi is made of:**

- Cellulose
- Chitin
- Silica
- Peptidoglycan

**5. A lichen is an association of:**

- Alga and fungus
- Bacterium and fungus
- Alga and bacterium
- Fungus and root

Answers: 1-R.H. Whittaker · 2-Mycoplasma · 3-Methane · 4-Chitin · 5-Alga and fungus